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Odd Semester Mid Term Examination, Academic Session: Jul-Nov 2026

FoSTA, School of Computer Science and Engineering

Department of Artificial Intelligence and Machine Learning

B.Tech. CSE-AIML

Course Code: AIM3120

Course: Computer Networks

Semester: V

Time: 1.5 hrs.

Max. Marks: 30

Instructions: All questions are compulsory.

Missing data, if any, may be assumed suitably.

Calculator is allowed.

SECTION A

S.No.		Marks	CO
Q.1	A data frame of 1011011 is to be transmitted using even parity. Determine the parity bit and write the complete transmitted frame. [2 Marks]		CO1
Q.2	A bit stream 1011010011 is transmitted at a data rate of 2 Mbps using the following line coding techniques: (a) Manchester Encoding (b) Differential Manchester Draw the corresponding line-coding waveforms for both techniques and clearly indicate the bit intervals.		CO1
Q.3	A digital communication system uses 4B/5B block coding to transmit data at a rate of 80 Mbps. Determine: 1. The number of data bits represented by each block. 2. The number of coded bits transmitted for every block. 3. The resulting transmission rate after 4B/5B encoding. 4. The percentage increase in the required bandwidth.		CO1
		2*4=8	
Q.4	Match the terms in Column A with the most appropriate descriptions in Column B .		
	Column A	Column B	
	1. Manchester Encoding	A. Reliable delivery using acknowledgements and retransmission	
	2. CRC	B. Line coding technique using transitions within each bit	
	3. Stop-and-Wait ARQ	C. Error-detection technique based on polynomial division	
	4. CSMA/CD	D. Medium access technique used to detect collisions	
SECTION B			

SECTION B

Q.5	A communication channel has a bandwidth of 6 kHz and an SNR of 24 dB. (i) Calculate the maximum channel capacity using Shannon's theorem. (ii) Calculate the Nyquist bit rate for binary signaling and compare it with the Shannon capacity.	3+4+4+4=15	C
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A shared communication channel operates at 1 Mbps and transmits 500-bit frames. The network generates an average traffic of 800 frames per second.

- Q.6/
- 1/ Calculate the offered load (G) and determine the throughput using Pure ALOHA. [1 Mark]
 - 2/ Calculate the throughput using Slotted ALOHA. [1 Mark]
 - 3/ Determine the percentage improvement in throughput achieved by Slotted ALOHA over Pure ALOHA. [1 Mark]
 4. State the maximum theoretical throughput of Pure ALOHA and Slotted ALOHA. [1 Mark]

CO2

A data link operates at 4 Mbps. Each frame contains 4000 bits, and the one-way propagation delay is 15 ms. A Sliding Window Protocol with a window size of 10 is used.

- Q.7/
1. Calculate the transmission time of one frame. [1 Mark]
 - 2/ Calculate the round-trip propagation delay. [1 Mark]
 - 3/ Calculate the channel utilization using the Sliding Window Protocol. [1 Mark]
 - 4/ Determine the minimum window size required to achieve 100% channel utilization. [1 Mark]

CO2

A 7-bit Hamming code uses even parity. The received codeword is:

0110011

- Q.8
1. Identify the positions of the parity bits. (1 mark)
 2. Calculate the parity-check values P_1, P_2 , and P_4 . (1 marks)
 3. Determine whether an error has occurred. If yes, identify the error bit position. (1 mark)
 4. Correct the received codeword and extract the original data bits. (1 mark)

CO2

SECTION-C

A company has been assigned the network block 192.168.50.0/24. The network administrator needs to divide this network among the following departments using VLSM (Variable Length Subnet Masking):

Department	Required Hosts
Department A	100
Department B	50
Department C	25
Department D	10

Q.9

Using VLSM, design an efficient subnetting scheme and answer the following:

1. Determine the appropriate subnet prefix/subnet mask required for each department based on its host requirement. [2 Mark]
2. Assign the Network Address and subnet range to each department, starting from the given network block. [2 Marks]
3. For Department B and Department C, determine the First Usable IP Address, Last Usable IP Address, and Broadcast Address. [2 Marks]
4. Calculate the total number of IP addresses allocated and the number of unused IP addresses remaining in the original /24 network block. [1 Mark]

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CO3